

Rahul Anant Ramane

Senior Software Engineer | GenAI Developer & LLM Solutions

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Professional Summary

GenAI engineer at Capgemini specializing in **LLM-powered application development**, RAG pipelines, multi-agent orchestration, and prompt engineering for enterprise clients. Delivered upto **80% productivity gains** across banking, insurance, and fintech engagements by designing intelligent automation agents using GitHub Copilot, GitLab Duo, LangChain/LangGraph, and open-source LLMs (Mistral, Llama). Proficient in Python, Django, REST APIs, and end-to-end GenAI solution architecture from prototype to production.

Technical Skills

AI & LLM: LLM APIs (GPT, Claude, Mistral, Llama), RAG, Prompt Engineering, Fine-Tuning Concepts, Agents, MCP, LangChain, LangGraph, GitHub Copilot, GitLab Duo, Continue Plugin

Vector DBs & Retrieval: FAISS, Embeddings, Semantic Search

Languages & Backend: Python, Java, DSA, Database, Django, FastAPI, REST APIs, Streamlit

Test Automation: Selenium, Playwright, Cucumber, TestNG, Tosca, REST-Assured, Postman

Certifications: IBM Python for Data Science · Generative AI · Prompt Engineering

Experience

Senior Software Engineer | Capgemini India

Sept 2024 – Present

– TOSCA Test Case Migration Agent

Internal

Python, Django, LLM Pipeline, REST APIs, Postman, Prompt Engineering

Collaborated with business stakeholders to design and build an LLM-powered Django backend with REST APIs that converts Tosca scripts into structured manual test cases at scale, validated via Postman; replaced a fully manual documentation process. Reached **97–100% conversion accuracy**, cutting manual effort by **80%**.

– Manual Test Case-to-Automation Script Generation

Pension Funds

GitHub Copilot Custom Agents, Java, Selenium, Azure DevOps

Enabled custom Copilot agents to pull test cases directly from Azure and auto-generate framework-aligned Selenium scripts, and test data removing a manual handoff step from the QA workflow. Achieved **87% accuracy** and cut effort by **75%**.

– Tosca → Playwright Script Generation

HSBC

Tosca Cloud, Tosca MCP, GitHub Copilot, Playwright

Designed and shipped custom GitHub Copilot agents that auto-convert Tosca test cases into executable Playwright scripts, eliminating manual rewrite effort across the QA team. Achieved **85% generation accuracy** with a **70% reduction** in scripting time.

– UiPath Validation & Tosca-to-Manual Test Case Migration

Nat West

Python, GenAI, XML Processing, Prompt Engineering, GitLab Duo

Built two IDE-integrated GitLab Duo solutions: a hybrid validation engine for UiPath QA reports (**70% accuracy, 50% effort cut**) and a Tosca-to-manual test case generator (**95–97% accuracy, 25% time savings**).

– Java Selenium & API Automation Framework

BNP Paribas

Continue Plugin, GPT OSS, Mistral, DOM/XPath, Java, Selenium, REST-Assured, ALM

Architected an AI-compatible REST-Assured and Selenium automation framework for a secure banking environment, using DOM-aware prompt templates to generate XPaths. Delivered **80–85% accuracy** and a **40% reduction** in per-test-case time.

– JMeter Script Generation Platform

Internal

Django, Streamlit, LLM, JMeter, Prompt Engineering, Performance Testing

Built a Django-backed Streamlit platform that auto-generates JMeter performance test scripts from natural-language prompts containing endpoint URLs and request details, standardizing output and removing manual scripting effort across teams.

Projects

QAgent — Multi-Agent RAG-Powered QA Automation Pipeline

Python · LangGraph · FastAPI · Next.js · Playwright · FAISS (Vector DB) · WebSockets

[Link](#)

- Architected a fault-tolerant multi-agent RAG pipeline using FAISS for semantic retrieval that autonomously converts business requirements into executable tests, with LLM fallback chain and real-time token streaming.

PathForge — AI-Powered Learning Roadmap Generator

TypeScript · Next.js 16 · React 19 · Drizzle ORM · Neon Postgres · Llama 3.1 · Three.js

[Link](#)

- Built AI-driven platform generating personalized learning roadmaps with interactive 3D graph visualization and a highly available Llama-3.1 inference engine.

Achievements

WOW Award: Recognized by Capgemini for excellent work in driving GenAI enablement across client engagements.

Automation Testing Certified: Intermediate level Ocean Test Certified

Education

B.Tech in Electronics & Telecommunication | PCCOE Pune, SPPU

2024